

Is the quasar PG1302-102 periodic?

NOAH KREVER,¹ MARIA CHARISI,² DANIEL J. D’ORAZIO,³ AND ZOLTÁN HAIMAN⁴

¹*Department of Astronomy, Columbia University, New York, NY 10027, USA; ndk2115@columbia.edu*

²*TAPIR, California Institute of Technology, Pasadena, CA 91125, USA; mcharisi@caltech.edu*

³*Department of Astronomy, Harvard University, 60 Garden Street, Cambridge, MA 02138, USA; daniel.dorazio@cfa.harvard.edu*

⁴*Department of Astronomy, Columbia University, New York, NY 10027, USA; zoltan@astro.columbia.edu*

ABSTRACT

The light-curve of the bright redshift $z \approx 0.3$ quasar PG1302-102 (hereafter PG1302) appears periodic at optical wavelengths (Graham et al. 2015; Charisi et al. 2015; D’Orazio et al. 2015), with sparse UV data consistent with the same ~ 5 yr cycle. The original evidence was based on a ~ 20 yr baseline combining CRTS, LINEAR, and archival data. Liu et al. (2018) argued that adding ~ 5 yr of ASAS-SN data removes the preference for a periodic model, but their analysis excluded ~ 11 yr of LINEAR and archival photometry. Using the full baseline, we compare DRW and DRW+sinusoid models and find that periodicity remains strongly favored. We introduce a Bayesian statistic to quantify changes in periodic evidence between baselines and compare PG1302 to a large ensemble of stochastic DRW realizations. From nearly 2×10^7 simulated DRW light-curves, only ~ 1 in 8×10^4 reproduce PG1302’s pre-ASAS-SN Lomb–Scargle peak, yielding 122 highly periodic DRW curves for comparison. For PG1302, the Bayesian Information Criterion difference is $\Delta\text{BIC} \approx 40$ on the LINEAR+CRTS baseline and ≈ 20 on the full baseline, continuing to favor a sinusoid despite a reduction of $\Delta(\Delta\text{BIC}) \approx 20$. None of the 122 DRW realizations reach comparably large ΔBIC values, and PG1302’s decrease in periodicity exceeds that of $\sim 95\%$ of the DRW sample. These results indicate that PG1302’s behavior is more consistent with a sinusoid-plus-noise model than with pure DRW variability. Within a DRW framework and assuming accurate photometric uncertainties, PG1302 remains a viable supermassive black hole binary candidate despite the reduced periodicity in the extended baseline.

Keywords: quasars: individual (PG1302-102) — quasars: supermassive black holes

1. INTRODUCTION

PG1302 attracted attention as a bright quasar with a periodic optical light-curve, which could be caused by the presence of a supermassive black hole binary (SMBHB). The periodicity was first claimed by Graham et al. (2015) based on a light curve which included data from: (i) the Catalina Real-time Transient Survey (CRTS; Drake et al. 2009); (ii) the Lincoln Near-Earth Asteroid Research (LINEAR; Sesar et al. 2011) program; (iii) the All Sky Automated Surveys (ASAS; Pojmanski 1997) as well as (iv) other, sparser archival data (Garcia et al. 1999; Eggers et al. 2000), providing a total baseline of ~ 20 yr.

Vaughan et al. (2016) cautioned that our incomplete knowledge of quasar variability, in combination with the limited quality of the optical light curves can lead to false detections of periodicity. This was demonstrated in the case of the quasar PSO J3334.2028+01.4075 (Liu et al. 2015), whose follow-up monitoring failed to show persistent periodicity (Liu et al. 2016).

Liu et al. (2018) re-analyzed PG1302’s light-curve with the addition of ASAS-SN data and performed a Bayesian model comparison, arguing that periodicity is no longer favored based on BIC differences.

Here we critically examine this claim by performing an independent likelihood analysis using the full optical dataset, DRW noise modeling, and a Bayesian framework. We find that the (DRW+sinusoid) model remains significantly favored over pure DRW when the full baseline is included.

We caution that these conclusions depend on the adopted noise model. If the true stochastic variability exhibits larger amplitude fluctuations on $\sim$ yr timescales than DRW models predict, apparent periodicity could emerge by chance.

2. SUMMARY OF PG1302-102’S LIGHTCURVE

Graham et al. (2015) first reported the existence of a quasar identified in the CRTS (Catalina Real-time Transient Survey) demonstrating periodicity, with a light

curve that can be fit to a sinusoid with a period of $P = 1884 \pm 88$ days and an amplitude $A = .14\text{mag}$ over a roughly 9 year period of increased observational frequency. As G15 notes, sparser historical data from CRTS dating back to 1993 is consistent with data from the more recent nine year period. The Lincoln Near-Earth Asteroid Research (LINEAR) data on PG1302, which dates back .5 cycles before CRTS data, is consistent with this sinusoidal fit. Liu et al. (2018) introduces analysis of more recent, publicly available extended light curve data from PG1302 observed in the All-Sky Automated Survey for Supernovae (ASAS-SN). The ASAS-SN data acquired spans from February of 2012 to March of 2018. L18 argues that the periodicity in ASAS-SN is out of phase with the historical LINEAR+CRTS data, with longer amplitude and period suggested by the combined, longer baseline.

L18 proceeded to bin the data from CRTS+LINEAR and ASASSN, and correct a 400 MJD overlap by offsetting the ASAS-SN light curve by the average difference of .17 mag, and binning correspondingly. We decide to use the ASAS-SN data as a pure extension of the LINEAR+CRTS baseline without overlap. As a result, the entire baseline we use is comprised of LINEAR+CRTS+ASASSN, however in proceeding to experiments and simulations, we keep in mind the pre and post ASAS-SN baselines as the two observational timeframes in which we will compare evidence of periodicity in PG1302 and corresponding simulated light curves.

For the combined LINEAR+CRTS (L+C) baseline, the Lomb–Scargle (LS) periodogram of PG1302 peaks at a period of ~ 5 yr, with a maximum normalized power of ≈ 0.83 . When the ASAS-SN data are added to form the full LINEAR+CRTS+ASAS-SN (L+C+A) light-curve, the LS peak shifts slightly and its height decreases to ≈ 0.76 , but remains at a period consistent with the original claim. These values motivate our choice of LS threshold when selecting DRW simulations that exhibit apparent periodicity similar to PG1302.

3. METHODOLOGY (FULL PAGE ISH)

3.1. Model framework

In general, observed light curves consist of $x = s + n$, where x =data, s =signal n =noise. The observed light-curve can be characterized by its covariance, which can be modeled as a sum of the signal and noise covariance.

$$C_x = C_s + C_n \quad (1)$$

where C_x is the observed data, $C_s = \langle s_i, s_j \rangle$ and $C_n = \langle n_i, n_j \rangle$ where i, j are elements of the total N elements in the light curve. The signal matrix's elements can be

expressed via the autocorrelation function

$$\langle s_i, s_j \rangle = \int_{-\infty}^{\infty} P(f) \cos(2\pi f \Delta t) df \quad (2)$$

where $P(f)$ is the power spectral density (PSD) of the signal, and Δt is the time lag between s_i and s_j .

In assessing the behavior of PG1302's optical variability, we must consider the central components of fit, namely, sinusoidal variation, stochastic variability, and photo-metric error. Two central frameworks for fitting quasar light-curves according to these components have emerged in recent years, namely, the damped-random-walk (DRW) and DRW + Sinusoid models. The DRW process is a stochastic process with interspersed random "walks" defined by a covariance matrix, and is frequently used today to model the optical variability in quasar light curves. Under the DRW model framework, the PSD in (2) would take the form

$$P(f) = \frac{2\sigma^2\tau^2}{1 + (2\pi\tau f)^2} \quad (3)$$

where σ^2 is short timescale variance and τ is the characteristic timescale. The DRW + Sinusoid model additively injects a sinusoid with the following form

$$A \sin(\omega t + t_0) \quad (4)$$

directly into the DRW signal. A, ω, t_0 are sinusoid amplitude, radial frequency, and phase shift (in days), respectively, with t being the current observation's MJD. Recent literature suggests that the optical variability of SMBHB candidates is best adequately modeled by such a combined framework.

The likelihood function for DRW-based models follows Kozłowski et al. (2017):

$$P(\mathbf{y}|\mathbf{p}) \propto |S + N|^{-1/2} |L^T C^{-1} L|^{-1/2} \exp\left(-\frac{\mathbf{y}^T C_{\perp}^{-1} \mathbf{y}}{2}\right), \quad (5)$$

where S is the signal covariance, N the noise covariance, L the design matrix encoding linear trends, and $C_{\perp}$ the projection matrix orthogonal to L .

Given likelihoods for both DRW and DRW+Sinusoid models, we compare their Bayesian Information Criteria,

$$\text{BIC} = k \ln N - 2 \ln \hat{L}, \quad (6)$$

where k is the number of parameters, N the number of observations, and $\hat{L}$ the maximum likelihood. Differences ΔBIC quantify the relative evidence for periodicity.

With likelihood functions defined for both the DRW and DRW+Sinusoid models, we compute their respective Bayesian Information Criteria (BICs) for each light

curve. This allows us to evaluate the relative evidence for periodicity through the quantity

$$\Delta\text{BIC} \equiv \text{BIC}_{\text{DRW}} - \text{BIC}_{\text{DRW}+\text{sin}}.$$

By comparing ΔBIC across different baseline lengths, we can test whether the reduction in periodicity evidence observed for PG1302 is consistent with that expected from a purely stochastic DRW process, or whether it is more consistent with a system containing a genuine sinusoidal component (as would arise in an SMBHB). Specifically, we assess whether the change

$$\Delta(\Delta\text{BIC}) = \Delta\text{BIC}_{\text{short}} - \Delta\text{BIC}_{\text{long}}$$

is systematically larger for pure DRW simulations than for PG1302, which would indicate that PG1302’s decrease in periodicity with additional data is inconsistent with DRW noise alone.

3.2. Simulations

To assess whether PG1302’s decrease in periodicity evidence with the addition of ASAS-SN data is consistent with pure stochastic variability, we generate large ensembles of purely DRW light-curves with the same sampling and photometric uncertainties as the real data.

We first construct simulated light-curves on the shorter LINEAR+CRTS (L+C) baseline by drawing DRW parameters (σ, τ) from broad uniform ranges and generating realizations using the covariance formulation described above. Each realization is sampled at the actual L+C observation times and perturbed with Gaussian noise using the corresponding magnitude errors. For every simulated L+C curve, we compute the Lomb–Scargle (LS) periodogram using the `astroML` implementation on a frequency grid, and record the location and height of the maximum peak.

As an initial test, we performed runs with 5×10^5 simulations and a threshold based on an earlier estimate of PG1302’s LS peak. After correcting the error bars and recomputing the LS periodogram for PG1302, we adopted the updated maximum power of ≈ 0.83 on the L+C baseline as our selection threshold. In a run of 10^6 L+C+ASAS-SN (L+C+A) simulations, only 12 realizations produced L+C peaks exceeding this value, corresponding to a success rate of ~ 1 in 8×10^4 curves.

We then scaled up to cluster runs. In a set of ten parallel runs, effectively 9×10^6 simulations, we obtained 112 L+C realizations with LS peaks above 0.83. An additional 10^7 simulations yielded 133 further successes, for a total of 245 LS-selected DRW curves from 1.9×10^7 simulations, again with a success rate of order $1/8 \times 10^4$.

To better match PG1302, we impose an additional constraint on the best-fit LS frequency. Using the L+C

LS periodogram of PG1302, we estimate the full width at half maximum (FWHM) of the main peak and translate this into an allowed period range of approximately 4.4–6.3 yr. We retain only those simulated L+C curves whose maximum LS peak both exceeds 0.83 and falls within this FWHM period window. Applying this filter to the initial 245 LS-selected curves leaves 40 realizations.

We then perform additional cluster runs totaling a further 3×10^7 simulated DRW light-curves, applying the same LS amplitude and FWHM frequency criteria. This yields 93 more curves that satisfy both cuts. After a final small set of runs, we assemble a sample of 122 purely DRW light-curves that (i) are sampled like PG1302 on the L+C baseline, (ii) have LS peak heights at least as large as PG1302’s L+C peak, and (iii) have best-fit LS periods within the FWHM of PG1302’s main peak.

For each of these 122 DRW realizations, we extend the light-curve to the full L+C+A baseline by continuing the same DRW process and sampling it at the ASAS-SN observation times, again adding Gaussian noise with the corresponding errors. This produces, for each simulated source, a pair of light-curves: one on the L+C baseline and one on the extended L+C+A baseline. In the next stage of the analysis, we fit both DRW and DRW+sinusoid models to each of these light-curves, compute the corresponding BIC values, and compare their ΔBIC and $\Delta(\Delta\text{BIC})$ to those obtained for PG1302.

4. RESULTS

4.1. Confidence Building Measures

Before interpreting the results for PG1302 and the DRW simulations, we validate our DRW and DRW+sinusoid inference pipeline.

First, we test recovery of DRW parameters using the long-baseline experiment of Kozłowski et al. (2017). We simulate light-curves with a 20,000 day baseline, 2,000 data points, DRW timescale $\tau = 200$ days, and structure-function amplitude $\text{SF}_\infty = 0.20$ mag, and fit them with our DRW likelihood. We reproduce the known bias pattern in the recovered (σ, τ) parameters over multiple realizations, in agreement with their results.

Second, we replicate the OGLE and SDSS-style tests of experiment length versus decorrelation timescale. Following Kozłowski, we simulate two sets of light-curves with $\text{SF}_\infty = 0.20$ mag, one with SDSS-like sampling (~ 60 epochs over 8 yr) and one with OGLE-like sampling (~ 445 epochs over 8 yr), spanning a range of input ratios $\rho_{\text{inp}} = \tau_{\text{inp}}/t_{\text{exp}}$. Our input–output comparisons in (σ, τ) space reproduce the trends seen in

their Figure 2, including the degradation in parameter recovery for large ρ_{inp} .

We then repeat similar tests using PG1302-like sampling. Using the actual L+C and L+C+A cadences, we simulate DRW light-curves with a grid of input (σ, τ) values and compare the recovered parameters from both grid searches and MCMC. For both baselines, we find that the inferred σ and τ track the inputs with modest bias, consistent with expectations from the OGLE/SDSS tests, confirming that our sampling pattern does not introduce qualitatively new biases beyond those already characterized.

Next, we validate MCMC parameter recovery for the combined DRW+sinusoid model. We generate DRW+sinusoid light-curves on the full L+C+A baseline with fiducial parameters $(\sigma, \tau, A, P, t_0)$ (e.g., $\sigma \approx 0.11$, $\tau \approx 100$ days, $A \approx 0.1$ mag, $P \approx 1825$ days, and a chosen phase), and fit them with our DRW+sinusoid likelihood. Over ensembles of simulated curves, the posterior modes for the DRW parameters and sinusoid amplitude and period recover the input values with small systematic offsets. Overlaying the best-fit models on the simulated data confirms that the inferred parameters produce light-curves consistent with the underlying realizations, even in cases where the posteriors are skewed or multimodal.

Finally, we compare our implementation of the Kozłowski covariance-based likelihood with the χ^2 -based likelihood used by Liu et al. (2018) and Zhu et al. (2020). Using binned versions of the PG1302 light-curve and model parameters supplied by Liu, we evaluate both likelihoods. Our χ^2 implementation reproduces their reported log-likelihoods for DRW and DRW+sinusoid models to within a few units, while the Kozłowski likelihood yields much larger absolute log-likelihood values but preserves the relative ordering between models. This gives confidence that, although the absolute BIC values can be inflated under the Kozłowski likelihood, ΔBIC remains a consistent measure of relative evidence between DRW and DRW+sinusoid.

4.2. Results

We first apply our model-selection framework directly to PG1302. Using the χ^2 -based likelihood in the style of Liu et al. (2018), we compute BIC values for both the DRW and DRW+sinusoid models on the L+C baseline and on the full L+C+A baseline. On the shorter L+C baseline, we obtain a $\Delta\text{BIC} = \text{BIC}_{\text{DRW}} - \text{BIC}_{\text{DRW+sin}} \approx 40$, indicating very strong preference for the DRW+sinusoid model. On the extended L+C+A baseline, the corresponding value is $\Delta\text{BIC} \approx 20$, still well within the regime usually inter-

preted as strong evidence for the more complex model. The Kozłowski covariance-based likelihood yields larger absolute BIC values but comparable differences between the two models.

The change in evidence when extending the baseline can be summarized by

$$\Delta(\Delta\text{BIC}) = \Delta\text{BIC}_{\text{short}} - \Delta\text{BIC}_{\text{long}} \approx 20,$$

so that the inclusion of ASAS-SN data reduces, but does not eliminate, the preference for a periodic component. In other words, within both likelihood formulations, PG1302 remains more consistent with a DRW+sinusoid model than with a pure DRW, even after the baseline is extended.

We then compare these values to those obtained from the 122 purely DRW simulations selected to match PG1302’s LS peak height and period on the L+C baseline. For each simulated light-curve, we compute ΔBIC on both the L+C and L+C+A baselines and form the corresponding $\Delta(\Delta\text{BIC})$. The resulting distributions show that most purely DRW curves have ΔBIC values near zero at both baselines, with only modest preference for DRW+sinusoid arising from chance fluctuations. None of the simulations reach the combination of large ΔBIC values seen for PG1302 at both baseline lengths.

By contrast, PG1302’s $\Delta(\Delta\text{BIC})$ is more negative than that of $\sim 95\%$ of the simulated DRW light-curves, i.e., its drop in periodicity evidence with the addition of ASAS-SN is larger than is typical in our DRW ensemble. However, this comparison must be interpreted with care. The DRW simulations in our sample were selected to barely exceed PG1302’s LS peak on the shorter baseline, and their absolute ΔBIC values are much smaller than PG1302’s at both lengths. As a result, small absolute changes in ΔBIC for these weakly periodic curves can appear similar to, or smaller than, the larger change seen in PG1302, even though their overall Bayesian evidence for periodicity is much lower.

Taken together, these results suggest the following picture. Within a DRW noise model, PG1302 exhibits unusually strong evidence for a sinusoidal component compared to DRW realizations with similar LS peak properties. Extending the baseline with ASAS-SN data reduces this evidence but does not bring PG1302 into the regime populated by purely stochastic DRW light-curves. The periodogram-based selection metric alone is therefore not a reliable proxy for Bayesian evidence: DRW simulations that satisfy the LS threshold can still have much weaker ΔBIC than PG1302. Our BIC analysis indicates that PG1302 remains more consistent with a sinusoid-plus-noise model than with a pure DRW, even

accounting for the decrease in periodicity evidence with additional data.

4.3. Other Caveats

Stochastic noise has not been ignored. Several previous works (e.g., Vaughan et al. 2016; Liu et al. 2015; Liu et al. 2018) have emphasized that red-noise variability in quasars can mimic periodicity over limited baselines. This concern is valid and is fully addressed in our framework: both the DRW and DRW+sinusoid models explicitly incorporate stochastic variability through their covariance structure, and the likelihoods used here marginalize over these noise components rather than treating the photometry as white noise. In addition, the simulations used for model selection are themselves pure DRW realizations, ensuring that any comparison between PG1302 and noise-only light curves reflects the same underlying variability model. Thus, our statistical evaluation of periodicity always takes stochastic noise into account, and the comparisons are performed within a consistent noise-modeling framework.

The DRW case and the Vaughan et al. discrepancy. Vaughan et al. (2016) reported a much weaker preference for periodicity when fitting PG1302 using a DRW-based model that includes an additional “error rescaling” parameter. In their formulation, the photometric errors are multiplied by a free factor ν , allowing the model to downweight small-scale variability by inflating the noise level. Introducing this extra parameter increases the flexibility of the DRW model and systematically pushes the best-fit damping timescale τ to much larger values. A long-timescale DRW produces smoother variability and can partially absorb low-amplitude sinusoidal features, reducing the apparent significance of periodicity. By contrast, our likelihood functions use the quoted measurement errors without such rescaling, leading to best-fit DRW parameters consistent with the short-timescale variability seen in the data. The resulting difference in inferred τ explains much of the discrepancy in periodicity evidence between our results and those of Vaughan et al. (2016). Ultimately, whether the small-timescale variability is intrinsic to the quasar or reflects underestimated errors remains an open question. A significantly larger error rescaling factor would reduce the evidence for periodicity even in our framework.

Uncertainty in the noise model. While the DRW model is widely used and provides a good empirical description of quasar optical variability on month-to-year timescales, it is not guaranteed to be the correct or complete model. Several studies have shown that quasar power spectra can deviate from the DRW

form, exhibiting either steeper slopes at high frequencies or additional structure on multi-year timescales. If PG1302’s true variability includes larger amplitude fluctuations on $\sim$ year baselines than predicted by the best-fit DRW, then extended observations may naturally produce apparent coherence or quasi-periodic trends without requiring a periodic component. In this case, the significance of the sinusoid would be overestimated. Thus, conclusions regarding PG1302’s periodicity necessarily depend on the adopted noise prescription. A more flexible stochastic model—e.g., a broken-power-law PSD or a continuous-time autoregressive moving average (CARMA) process—may yield different relative evidence for periodicity, though assessing such models is beyond the scope of this work. We therefore emphasize that our conclusions apply within the DRW framework and that any deviation from DRW-like behavior could alter the inferred strength of periodicity in PG1302.

5. CONCLUSIONS

Within the framework of a damped random walk (DRW) noise model and assuming that the quoted photometric uncertainties are reliable, our analysis finds that PG1302 continues to exhibit statistically significant evidence for a periodic component. Using both covariance-based and χ^2 -based likelihoods, the DRW+sinusoid model remains strongly preferred over a pure DRW description on the original LINEAR+CRTS (L+C) baseline, and this preference persists, albeit at reduced significance, when the ASAS-SN data are included to form the full L+C+ASAS-SN (L+C+A) baseline. The drop in ΔBIC with the extended baseline is substantial but does not erase the overall preference for a sinusoidal contribution.

By constructing large ensembles of purely DRW light curves with PG1302-like sampling and selecting realizations that match PG1302’s Lomb–Scargle peak height and period on the L+C baseline, we find that such strongly and coherently periodic DRW fluctuations are rare. Moreover, even among this carefully selected subset, the resulting ΔBIC values are generally much smaller than those obtained for PG1302 at both baseline lengths. PG1302 thus occupies an atypical region of parameter space compared to purely stochastic DRW realizations: its Bayesian evidence for a sinusoid remains unusually strong, and its decrease in periodicity evidence with added data does not resemble the behavior of the simulated DRW ensemble.

At the same time, our conclusions are conditional on the assumed form of the stochastic variability and the accuracy of the photometric error estimates. If the true noise process has larger amplitude fluctuations on multi-

year timescales than predicted by the best-fit DRW, or if the measurement uncertainties are significantly underestimated, then a DRW-like process could more easily generate apparent periodicity over the existing baseline. In that regime, the significance of the sinusoidal component would be reduced, and PG1302's apparent periodicity could plausibly be attributed to random chance. The distinction between these possibilities is ultimately limited by the baseline length, sampling, and the fidelity of the noise model.

Further progress can be made along at least two complementary directions. First, extended optical monitoring of PG1302 over additional cycles is essential. If the signal arises from a genuine supermassive black hole binary, the phase and amplitude of the variability should remain coherent over significantly longer baselines than currently available. A decisive outcome is likely to require monitoring over at least one to several additional

cycles, corresponding to a timescale of order 10–20 yr. Second, independent observational probes could help break degeneracies inherent in the optical light-curve alone. Multiwavelength variability, particularly in the UV and X-ray bands, may reveal signatures expected from a relativistic Doppler boost or from changes in the accretion flow. Likewise, time-resolved spectroscopy of the broad emission lines could provide dynamical or structural evidence for binary motion or disk precession. Together, longer-baseline photometric monitoring and complementary multiwavelength and spectroscopic observations offer a path toward determining whether PG1302 is best described as a rare

Financial support was provided by NASA through grants NNX15AB19G, NNX17AL82G and 16-SWIFT16-0015, and by NSF grant 1715661.

REFERENCES

- Charisi, M., Bartos, I., Haiman, Z., Price-Whelan, A. M., & Márka, S. 2015, *MNRAS*, 454, L21
- D’Orazio, D. J., Haiman, Z., & Schiminovich, D. 2015, *Nature*, 525, 351
- Drake, A. J., Djorgovski, S. G., Mahabal, A., et al. 2009, *ApJ*, 696, 870
- Eggers, D., Shaffer, D. B., & Weistrop, D. 2000, *AJ*, 119, 460
- Garcia, A., Sodré, L., Jablonski, F. J., & Terlevich, R. J. 1999, *MNRAS*, 309, 803
- Graham, M. J., Djorgovski, S. G., Stern, D., et al. 2015, *Nature*, 518, 74
- Liu, T., Gezari, S., & Miller, M. C. 2018, *ApJL*, submitted, e-print arXiv:1803.05448, arXiv:1803.05448
- Liu, T., Gezari, S., Heinis, S., et al. 2015, *ApJL*, 803, L16
- Liu, T., Gezari, S., Burgett, W., et al. 2016, *ApJ*, 833, 6
- Pojmanski, G. 1997, *AcA*, 47, 467
- Sesar, B., Stuart, J. S., Ivezić, Ž., et al. 2011, *AJ*, 142, 190
- Vaughan, S., Uttley, P., Markowitz, A. G., et al. 2016, *MNRAS*, 461, 3145